EX 6:

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

PROGRAM

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix


iris = load_iris()


X = iris.data

y = iris.target


X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)


# Naive Bayes

model = GaussianNB()

model.fit(X_train, y_train)


# Prediction

y_pred = model.predict(X_test)


print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred))
```

print("\nAccuracy:")

print(accuracy_score(y_test, y_pred))

OUTPUT :

```
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Accuracy:
1.0
```